2021-Summer Paper 2 – Urology

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Q1. Describe factors affecting outcome of ESWL. Discuss complications of ESWL and strategies to avoid them.

Type: LAQ (25 marks) (Mechanism, indications and current status are answered in 2019-Summer Paper 2, Q2.)

Introduction

- ESWL (extracorporeal shock wave lithotripsy, now SWL) breaks stones with shock waves (peak 30–100 MPa, negative tail about −10 MPa) generated at F1 and focused on the stone at F2.
- Stones break by spallation, shear, squeezing and cavitation. **Cavitation** also causes most tissue injury, so steps that improve outcome usually reduce complications too.
- Outcome = SFR (stone-free rate) and need for retreatment.

Factors affecting outcome

1. Stone factors

Factor	Favourable	Unfavourable
Size (most important)	<10 mm	Renal >20 mm (PCNL [percutaneous nephrolithotomy] first); ureteric ≥10 mm favours URS (ureteroscopy)
Density on NCCT (non-contrast computed tomography)	<900 HU (Hounsfield units)	>1000 HU
Composition	Calcium oxalate dihydrate, uric acid	Cystine, brushite, calcium oxalate monohydrate (dense, homogeneous)
Impaction (ureter)	Recent, mobile	Chronic impaction (soft-tissue ring of thick ureteral wall on CT, unchanged position): may be the strongest predictor
Burden and site	Single, renal pelvis or upper calyx	Multiple sites; lower pole >1 cm

- Infection stones and stones in several sites: prefer a more efficient modality.

2. Renal anatomical factors

- **Lower pole:** fragments drain against gravity. **Lower Pole I (Albala, 2001):** stones >1 cm, SFR 21% SWL vs 91% PCNL. A steep infundibulopelvic angle and long, narrow infundibulum reduce clearance.

- Obstruction distal to the stone (UPJO [ureteropelvic junction obstruction], infundibular stenosis, calyceal diverticulum) traps fragments. Horseshoe and ectopic kidneys drain poorly.

3. Patient factors

- **SSD (skin-to-stone distance) >10 cm** (obesity) predicts failure. Skeletal deformity can prevent targeting.
- Respiratory movement of 10–15 mm moves the stone out of F2.
- **Triple D score:** stone density, stone volume and skin-to-stone distance together predict success.

4. Machine and technique factors

- **Energy source:** electrohydraulic (HM3) breaks best; piezoelectric has low energy and a tiny focal zone; modern machines are electromagnetic. A focal zone wider than the stone breaks better.
- **Coupling:** air pockets in the gel cut fragmentation by up to two-fifths.
- **Rate 60–90 shocks/min** beats 120/min, because the bubble cloud clears between shocks. **Ramping** energy up also helps.
- **Analgesia:** GA (general anaesthesia) or sedation keeps the stone in focus.
- Positioning (prone for mid-ureteric stones over the sacrum), continuous imaging, experienced team. Little gain beyond two sessions.
- **Adjuncts:** α-blockers (faster fragment clearance); potassium citrate; percussion, diuresis and inversion for lower pole fragments.

Complications

Group	Complication	Comment
Fragment-related	Renal colic; **steinstrasse** ("stone street" of fragments in the ureter); residual fragments	Steinstrasse rises several-fold with burden >2 cm^2
Infective	UTI (urinary tract infection), pyelonephritis, urosepsis	Infection stones, stents
Renal tissue injury	Haematuria (transient, common); **subcapsular or perirenal haematoma** (minority; rarely embolisation)	Risk: age, hypertension, obesity, diabetes, coronary disease, thrombocytopenia, high energy, many shocks, narrow-focus machines
Other organs	Liver, spleen, pancreas, colon, lung (children), skeletal muscle	Very rare
Cardiac	Arrhythmia during treatment	Pause or ECG (electrocardiogram) gating
Skin	Petechiae, bruising	Minor

- **Refuted long-term effects:** new hypertension, diabetes, decline in renal function and infertility.

Strategies to avoid complications

1. **Select.** Absolute contraindications: pregnancy; uncorrected coagulopathy or ongoing antithrombotics; untreated UTI; obstruction distal to the stone; aneurysm near the stone (aortic >4 cm); severe uncontrolled hypertension. Send large, dense, lower pole or impacted stones to URS or PCNL.
2. **Prepare.** Urine culture; treat bacteriuria. Prophylaxis only with risk factors (stent, nephrostomy, struvite, recent UTI, bowel reconstruction). Stop antithrombotics for the day and 48 hours after. Control blood pressure; reprogramme pacemakers.

3. **Protect the kidney.** Slow rate; ramp over at least 4 minutes (induces protective renal vasoconstriction); fewest shocks and lowest effective energy.
4. **Prevent steinstrasse.** PCNL or URS for large burdens. Routine stenting does not improve SFR, though it may reduce steinstrasse with large stones. α-blockers after SWL.
5. **Treat early.** Steinstrasse: MET (medical expulsive therapy), SWL to the lead fragment or URS; **nephrostomy if fever or obstruction**. Haematoma: bed rest, transfusion, embolisation if bleeding continues.
6. **Follow up** with KUB (kidney-ureter-bladder radiograph) and ultrasound.

Recent advances

- **Burst wave lithotripsy** (ultrasound bursts at 4–8 MPa: less cavitation, no anaesthesia, uniform fragments); **ultrasonic propulsion** of fragments.

Diagram to draw: ellipsoid reflector with F1, F2, stone and gel coupling; box of failure predictors.

> **Key points to remember:**
> - Size is the main predictor; HU >1000 and SSD >10 cm predict failure.
> - Lower Pole I: SFR 21% SWL vs 91% PCNL for lower pole stones >1 cm.
> - Slow rate (60–90/min), ramping and good coupling improve breakage and reduce injury.
> - Steinstrasse risk rises with burden >2 cm^2; no routine stent.

Q2. Describe selection criteria for cadaver renal donor. Describe in details kidney retrieval surgery.

Type: LAQ (25 marks) (Marginal donor in 2016-Summer Paper 3, Q3f; brain death certification in 2018-Summer Paper 4, Q3c.)

Introduction

- A cadaver (deceased) donor is **DBD** (donation after brain death: circulation maintained until cold perfusion) or **DCD** (donation after circulatory death: warm ischaemia).
- India: **THOA (Transplantation of Human Organs Act) 1994**, amended 2011, Rules 2014, recognises brain-stem death; **NOTTO** (National Organ and Tissue Transplant Organisation) allocates organs.

Selection criteria

1. Legal death and consent

- **DBD:** apnoeic coma of known irreversible cause; exclude drugs, hypothermia, metabolic causes. All brainstem reflexes absent (pupillary, corneal, oculocephalic, oculovestibular, cranial motor response, gag and cough). **Apnoea test:** no breathing effort at $PaCO_2$ (arterial carbon dioxide tension) ≥60 mmHg.
- India: **4-member Board** (hospital in-charge, independent specialist, neurologist or neurosurgeon, treating doctor; none from the transplant team); **two tests at least 6 hours apart**; Form 10.
- **DCD (Maastricht):** I dead on arrival; II failed resuscitation (both uncontrolled); **III arrest after withdrawal of support (controlled, commonest)**; IV arrest in a brain-dead donor; V euthanasia.
- Next-of-kin consent after a **mandatory request** by ICU (intensive care unit) staff.

2. Donor evaluation

- Cause of death, medical and social (high-risk behaviour) history; kidney: terminal creatinine, urine output, hypertension, diabetes, ultrasound.
- Laboratory: ABO and HLA (human leucocyte antigen) typing; serology and NAT (nucleic acid testing) for HIV (human immunodeficiency virus), hepatitis B and C, CMV (cytomegalovirus), EBV (Epstein–Barr virus), syphilis, toxoplasma.

3. Absolute contraindications

- Active or metastatic cancer, haematological malignancy, history of melanoma, untreatable viral infection.
- Treated bacterial infection (even bacterial meningitis) is acceptable; viral encephalitis needs care.

4. Standard versus marginal donor

Category	Definition
ECD (expanded criteria donor), UNOS (United Network for Organ Sharing)	Age ≥60, or 50–59 with 2 of: hypertension, creatinine >1.5 mg/dL, death from CVA (cerebrovascular accident)
KDPI (Kidney Donor Profile Index)	Replaced ECD in the US (2014); >85% = marginal; longevity matching with EPTS (Estimated Post-Transplant Survival)
Paediatric <15 kg	En bloc transplant

- **Remuzzi biopsy score** for marginal kidneys: 0–3 single, 4–6 dual, ≥7 discard.
- **Donor management (rule of 100s):** systolic pressure >100 mmHg, urine >100 mL/h, PaO_2 (arterial oxygen tension) >100 mmHg, haemoglobin >100 g/L. Vasopressin treats diabetes insipidus.

Kidney retrieval surgery (multi-organ, DBD)

1. **Checks:** death certificate, consent, blood group; coordinate thoracic, liver, pancreas and kidney teams.
2. **Incision:** midline xiphoid-to-pubis with median sternotomy.
3. **Exposure:** mobilise right colon and small-bowel mesentery off the retroperitoneum (Cattell–Braasch manoeuvre); expose aorta and IVC (inferior vena cava).
4. **Prepare cannulation:** encircle the distal aorta above the bifurcation; ligate the inferior mesenteric artery; prepare the supracoeliac aorta for clamping. **Look for aberrant renal arteries**; keep accessory arteries from the iliacs.
5. **Cold perfusion:** systemic heparin; cannulate the distal aorta; clamp the supracoeliac aorta; **vent** the IVC (or right atrium); flush cold UW (University of Wisconsin) or HTK (histidine–tryptophan–ketoglutarate) solution; crushed ice; open Gerota's fascia laterally to cool the kidneys.
6. **Remove heart, lungs, liver and pancreas first.**
7. **Kidneys en bloc:** mobilise both with perinephric fat off the posterior wall; keep periureteric tissue; divide ureters low. Divide aorta and IVC (upper limit set by the liver team, lower at the bifurcation); lift everything anteriorly and divide prevertebral tissue.
8. **Extras:** iliac vessels for reconstruction; spleen and lymph nodes for crossmatch.

DCD: after death and a no-touch period, **super-rapid** laparotomy, direct aortic cannulation and IVC venting limit warm ischaemia; kidneys usually go on machine perfusion.

Back-bench preparation

- Open the aorta posteriorly to see the renal ostia and divide it between them; keep the IVC with the right kidney to lengthen the short right renal vein.
- Keep an aortic **Carrel patch**; ligate tributaries and lymphatics (prevents lymphocele); trim fat without decapsulation; spare the lower pole–ureter triangle.
- Document anatomy and injuries; biopsy marginal kidneys; triple-bag in cold solution on ice; label.

Preservation

- Cooling to 4 °C slows metabolism. **Hypothermic machine perfusion** reduces DGF (delayed graft function) versus static cold storage (Moers, 2009), not long-term survival. Keep cold ischaemia short (thrombosis risk >24 hours).
- **Pitfalls:** polar artery or intimal injury, short right vein, ureteric devascularisation, capsular tears.
- **Also know:** HCV (hepatitis C virus)-viraemic donors with antivirals; normothermic perfusion (experimental).

Diagram to draw: aorta and IVC with distal cannula, supracoeliac clamp, IVC vent and en bloc kidneys.

> **Key points to remember:**
>
> - THOA: 4-member Board, two tests ≥6 h apart; apnoea test at $PaCO_2$ ≥60 mmHg.
> - ECD: ≥60, or 50–59 with 2 of hypertension, creatinine >1.5, CVA death; KDPI >85%.
> - Distal aortic cannula, supracoeliac clamp, IVC vent, cold UW/HTK and ice.
> - Kidneys en bloc last; Carrel patch, IVC for right vein; machine perfusion reduces DGF.

Q3a. Post prostatectomy incontinence.

Type: SAQ (10 marks)

Definition and incidence

- PPI (post-prostatectomy incontinence) is mainly **SUI (stress urinary incontinence)** after RP (radical prostatectomy), the commonest cause of male SUI. Rates vary with definition ("no pad" vs "safety pad"); expert no-pad continence is in the mid-90s % at 1 year.

Pathophysiology

- RP removes the internal sphincter (bladder neck, prostatic urethra), so continence depends on the external **rhabdosphincter**.
- Causes: apical injury to rhabdosphincter or pudendal branches, short membranous urethra, afferent denervation; also detrusor overactivity, low compliance, **VUAS (vesicourethral anastomotic stenosis)**.
- Nerve sparing and posterior reconstruction (Rocco stitch) give **earlier, not better final** continence.

Evaluation

- History (type, severity, radiotherapy, cancer status), bladder diary, **24-hour pad weight**: mild <200 g, moderate 200–400 g, severe >400 g.
- **Standing cough test**, grade 0 to 4 (leak reported but not seen; delayed drops; early drops; delayed stream; early persistent stream).
- Urinalysis, PVR (post-void residual), PSA (prostate-specific antigen). **Cystoscopy** excludes VUAS and judges residual sphincter coaptation; urodynamics when needed.

Management

1. **Wait 6–12 months** (improves up to 2 years). Supervised **PFMT (pelvic floor muscle training)** with biofeedback, ideally preoperative; pads, condom catheter or clamp.
2. **Duloxetine** off-label. **Bulking agents are not advised.**
3. **Surgery** (AUA/SUFU [American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction] 2019, EAU [European Association of Urology]):

Situation	Choice
Mild–moderate, no radiotherapy	**Male sling**: AdVance (repositions the bulbar urethra; needs a coapting sphincter) or adjustable systems (ATOMS [adjustable transobturator male system])
Severe or any radiotherapy	**AUS (artificial urinary sphincter), AMS 800**: gold standard; cuff 61–70 cm H_2O
ProACT balloons	Avoid after radiotherapy

- **MASTER trial:** male sling non-inferior to AUS at 12 months.
- AUS: social continence (0–1 pad) 55–77%; revision 10–29% at 3 years; lasts 8–10 years. Erosion risks: radiotherapy, prior erosion, urethroplasty (use a **transcorporal cuff**).

Diagram to draw: AMS 800 with cuff, scrotal pump and pressure-regulating balloon.

> **Key points to remember:**
> - Wait 6–12 months with PFMT before surgery.
> - Sling for mild–moderate; AUS for severe or irradiated.
> - MASTER: sling non-inferior to AUS at 12 months.

Q3b. Early versus delayed VVF repair.

Type: SAQ (10 marks)

Introduction

- VVF (vesicovaginal fistula): an epithelialised bladder–vagina communication causing continuous leakage.
- Causes: **obstetric** (over 90% in low-resource settings; obstructed labour) and **iatrogenic** (over 75% in high-resource settings, mainly hysterectomy); radiation; malignancy.
- Common mechanism: **ischaemic necrosis**, then a tract that epithelialises with time.
- Presents: cystotomy, days; thermal injury, 7–10 days; radiation, months to years.

The debate

	Early repair	Delayed repair
Definition (varies)	Days, <6 weeks or <3 months	Traditionally ≥3 months
Rationale	Clean iatrogenic tissue heals well; ends wetness and distress early	Inflammation settles, necrotic tissue sloughs, margins become clear
Evidence	**Comparable closure** in most series; Waaldijk: high first-attempt closure with immediate obstetric repair	One small series: more recurrence when repaired <6 weeks
Drawback	Friable tissue if inflamed	Prolonged distress, no better outcome

- Some authors **avoid 4–12 weeks** (active healing phase).
- **EAU (European Association of Urology) consensus:** no minimum delay if **no active infection and no recent irradiation**.

Recommended timing by cause

- **Clean post-hysterectomy:** early repair. **Obstetric with necrosis:** catheter trial, wait about 3 months.
- **Radiation fistula:** wait **≥12 months** after radiotherapy; biopsy for recurrence; interposition flap; diversion if the bladder is poor.

Conservative trial (during the wait)

- Catheter drainage cures about 15%: best if small (<1 cm, ideally <3 mm), started **within 3 weeks**, fresh and non-irradiated. Once epithelialised (≥6 weeks), unlikely. EAU: if tried, at least 12 weeks.
- **Also know:** fulguration of tiny tracts; fibrin glue.

Principles at repair

- Treat infection, optimise nutrition; ureteric stents if within 1 cm of a ureter.
- Vaginal or abdominal (open, laparoscopic, robotic): about 90% first-attempt closure. Tension-free, multilayer closure; **interposition flap** (Martius, omentum, peritoneum) if irradiated or recurrent; catheter often ≥2 weeks.

Diagram to draw: timeline: early window, 4–12 week zone, 3 months (obstetric), 12 months (radiation).

Key points to remember:

- Early repair for clean iatrogenic VVF; closure comparable to delayed.
- Delay ~3 months obstetric, ≥12 months radiation.
- Catheter trial works only if small, fresh and started within 3 weeks.

Q3c. Perineal urethrostomy.

Type: SAQ (10 marks)

Definition

- PU (perineal urethrostomy): the proximal bulbar urethra is opened onto perineal skin as a permanent or staged new meatus; the patient voids sitting. It **bypasses** rather than reconstructs a diseased anterior urethra.

Indications

- **Complex anterior strictures:** LS (lichen sclerosus) panurethral disease, multiple failed urethroplasties, failed hypospadias repair.
- **Elderly or high-risk patients** who do not want or cannot withstand staged reconstruction.
- **First stage** of a staged (Johanson-type) urethroplasty.
- After **penectomy** or distal urethrectomy for penile or urethral cancer.
- Temporary: stricture after phalloplasty; resectoscope access in morbid obesity; Fournier gangrene.
- AUA (American Urological Association) guideline (2016): PU is an acceptable long-term alternative to urethroplasty.
- LS spares the proximal urethra, so PU is easy in LS.

Contraindications and limits

- Disease reaching the proximal bulbar or membranous urethra; prior pelvic radiation and LS increase restricture.

Techniques

1. Classical (Blandy) PU: inverted-U perineal skin flap advanced into a ventral bulbar urethrotomy (a random-pattern advancement flap).

2. BMG (buccal mucosal graft) PU, steps:

1. High lithotomy; inverted-U incision; divide bulbospongiosus in the midline.
2. Free the spongiosum; **transect** it; **dorsal urethrotomy** until healthy mucosa.
3. Bring the healthy ventral urethra to perineal skin; calibrate proximally to 30 Fr; close the distal urethra.
4. Quilt a BMG on the corpora to complete the dorsal wall up to the skin.

- The graft prevents the circumferential scar that narrows a deep PU.

Complications and outcomes

- **Stenosis of the urethrostomy** (main failure): revision or periodic dilation.
- Spraying, dribbling, perineal wetness, wound infection, body-image concerns.
- Most succeed at mid-term with high satisfaction in selected patients.

Diagram to draw: sagittal perineum showing inverted-U flap sutured into the opened bulbar urethra.

> **Key points to remember:**
> - PU bypasses, rather than reconstructs, the diseased anterior urethra.
> - Ideal for LS, failed repairs and elderly patients; patient voids sitting.
> - BMG PU prevents circumferential scarring; calibrate to 30 Fr.

Q3d. Medical expulsive therapy in urinary calculi.

Type: SAQ (10 marks)

Definition

- MET (medical expulsive therapy): drugs that help a ureteric stone pass into the bladder. **α-blockers** are the main agents.

Rationale

- Spontaneous passage depends on **width**, size and site:

Width	Passage	Location	Passage
0–3 mm	98%	Vesicoureteric junction	92% (small stones)
5 mm	65%	Distal ureter	75%
>6.5 mm	9%	Proximal ureter	48%

Mechanism

- **α1A and α1D receptors** dominate in the distal ureter and VUJ (vesicoureteric junction). Blockade reduces spasm and peristaltic force and frequency, so a larger urine bolus pushes the stone.
- Calcium-channel blockers (nifedipine); PDE5 (phosphodiesterase-5) inhibitors under study. **NSAIDs (non-steroidal anti-inflammatory drugs) do not aid passage.**

Drugs

- **Tamsulosin 0.4 mg daily** (most studied); **silodosin 8 mg daily** (better expulsion in meta-analyses; twice the retrograde ejaculation); alfuzosin, doxazosin.

Evidence

Study	Result
SUSPEND (Pickard, 2015): 1136 patients, tamsulosin, nifedipine or placebo	No difference in need for intervention at 4 weeks (19% vs 20%)
STONE (Meltzer, 2018): stones <9 mm	Tamsulosin no better than placebo
Meta-analyses and a large distal-stone RCT (randomised controlled trial)	Benefit for **distal stones >5 mm**

- Stratified, the trials agree: benefit lies in **distal stones of 5–10 mm** (smaller pass anyway; larger rarely pass).

Indications and guideline position

- AUA (American Urological Association) and EAU (European Association of Urology): offer MET for **ureteric stones <10 mm**, especially distal, when immediate intervention is not needed (EAU: distal >5 mm).
- **After SWL (shock wave lithotripsy)** for faster fragment clearance. Children: off-label but effective.

Contraindications (need urgent drainage instead)

- Infection with obstruction or sepsis, solitary or bilateral obstruction, renal deterioration, refractory pain or vomiting.

Protocol

- NSAID analgesia; counsel on off-label use, dizziness, hypotension, retrograde ejaculation.
- Periodic KUB (kidney-ureter-bladder radiograph) or ultrasound; **intervene after 4–6 weeks** if not passed.

Diagram to draw: ureter with α1 receptors at the distal ureter and VUJ and a stone above it.

> **Key points to remember:**
>
> - Tamsulosin 0.4 mg; best for distal stones 5–10 mm.
> - SUSPEND 2015: no benefit in an unselected population.
> - Trial of passage 4–6 weeks, then intervene.

Q3e. TOT versus TVT in SUI.

Type: SAQ (10 marks) (TOT answered in 2020-Summer Paper 2, Q3f.)

Introduction

- Both are polypropylene **MUS (mid-urethral slings)** for SUI (stress urinary incontinence), based on the **integral theory** (Petros and Ulmsten, 1990): a loose mid-urethral tape replaces the pubourethral ligaments as a backboard on straining.
- **TVT (tension-free vaginal tape, retropubic):** Ulmsten, 1995–1998. **TOT (transobturator tape):** Delorme, 2001, to avoid bladder, bowel and vessel injury.

Technique differences

	TVT (retropubic)	TOT (transobturator)
Path	Vagina → retropubic space → suprapubic skin, hugging the pubis; cystoscopy essential	Labiocrural fold → obturator membrane → vagina (outside-in), or reverse (inside-out)
Structures at risk	Bladder, bowel, iliac and epigastric vessels	Obturator nerves and vessels, vaginal fornix

Outcomes

- **TOMUS (Richter, 2010):** 597 women; 12-month objective cure **80.8% retropubic vs 77.7% transobturator** (not different); voiding dysfunction needing surgery 2.7% vs 0%.
- TOMUS at 5 years: 51.3% vs 43.4%; equivalence not shown; satisfaction similar.
- **Cochrane (Ford, 2017):** cure similar up to and beyond 5 years; repeat surgery possibly commoner after TOT.

Complications (Cochrane, 2017)

Complication	TVT	TOT
Bladder perforation	**4.5%**	**0.6%**
Groin pain	1.3%	**6.4%**
Voiding dysfunction, blood loss, vascular injury	More	Less
Mesh exposure	About 2%	About 2%

- Urethral injury during either: **do not place mesh** (AUA [American Urological Association] guideline).

Choosing

- **TVT preferred: ISD (intrinsic sphincter deficiency)** (ALPP [abdominal leak point pressure] <60 cm H_2O or MUCP [maximum urethral closure pressure] <20 cm H_2O), fixed urethra, recurrent SUI. Failure is 12 times likelier after TOT with low leak point pressure (O'Connor, 2006).
- **TOT preferred:** previous retropubic surgery, obesity, avoiding bladder injury.
- AUA: either may be offered to the index patient.
- **Also know:** single-incision mini-slings (non-inferior to TOT); mesh warnings (FDA [Food and Drug Administration] 2010; UK pause 2018): counsel about alternatives (colposuspension, autologous sling, bulking).

Diagram to draw: coronal pelvis: U-shaped retropubic TVT versus flat TOT through the obturator foramina.

Key points to remember:

- TOMUS: 80.8% vs 77.7%, equivalent at 1 year.
- TOT: less bladder injury (0.6% vs 4.5%), more groin pain (6.4% vs 1.3%).
- Prefer retropubic TVT in ISD or fixed urethra.

Q3f. Hereditary urological malignancies.

Type: SAQ (10 marks)

Principles

- **Knudson two-hit hypothesis:** every cell carries one inherited mutant tumour-suppressor allele, so one more hit causes cancer: **early, bilateral, multifocal** tumours.
- **4–6% of RCC (renal cell carcinoma)** is familial; all familial RCC syndromes are **autosomal dominant**; only hereditary papillary RCC (HPRC) involves an activated oncogene.

Hereditary kidney cancer syndromes

Syndrome	Gene (chromosome)	Kidney tumour	Other features
VHL (von Hippel–Lindau)	VHL (3p25)	Clear cell, bilateral, multifocal (~50%)	Retinal angioma, CNS (central nervous system) haemangioblastoma, phaeochromocytoma, pancreatic cysts
HPRC	MET (7q31) oncogene	Multiple bilateral papillary (type 1-like)	None
HLRCC (hereditary leiomyomatosis and RCC)	FH (fumarate hydratase, 1q42–43)	**Aggressive**, often solitary papillary	Skin and uterine leiomyomas
BHD (Birt–Hogg–Dubé)	FLCN (folliculin, 17p11.2)	Chromophobe, hybrid oncocytic, oncocytoma	Fibrofolliculomas, lung cysts, pneumothorax
TSC (tuberous sclerosis complex)	TSC1 (9q34), TSC2 (16p13)	Multiple AMLs (angiomyolipomas); RCC 1–3%	Angiofibromas, cortical tubers

- **VHL type 1** (deletions, truncations): no phaeochromocytoma; **type 2** (missense): phaeochromocytoma.
- **Also know:** SDH (succinate dehydrogenase)-deficient RCC (paraganglioma); BAP1 (uveal melanoma, mesothelioma); Cowden (PTEN; breast, thyroid).

Management principles for kidney

- **3 cm rule:** watch tumours <3 cm; nephron-sparing surgery at ≥3 cm (VHL, HPRC, BHD) to save function for repeat operations. **Exception: HLRCC**: excise promptly, wide margins.
- TSC AML: everolimus; embolisation or partial nephrectomy if ≥5 cm. **Belzutifan** (HIF-2α [hypoxia-inducible factor 2α] inhibitor, FDA [Food and Drug Administration]-approved 2021) for VHL.

Other urological cancers

- **Prostate: BRCA2**, BRCA1, ATM, CHEK2, PALB2, HOXB13, mismatch repair genes; father plus brother: 9-fold risk. Earlier screening, cautious active surveillance, **PARP (poly ADP-ribose polymerase) inhibitors**.
- **UTUC (upper tract urothelial carcinoma): Lynch syndrome** (MLH1, MSH2, MSH6, PMS2): 14-fold risk, about 75-fold with MSH2. Suspect if age <60 or Lynch cancer history; MMR (mismatch repair) immunohistochemistry, then germline testing.
- **Testis:** affected brother RR (relative risk) 8–12, father 2–4.

- **Also know:** Wilms tumour (WT1 gene: WAGR [Wilms, aniridia, genitourinary anomalies, retardation] and Denys–Drash; Beckwith–Wiedemann).
- **Approach:** genetic counselling, germline testing, screening relatives.

Diagram to draw: VHL loss → HIF accumulation → VEGF (vascular endothelial growth factor) → angiogenesis.

> **Key points to remember:**
> - All familial RCC is autosomal dominant; 3 cm rule except HLRCC.
> - VHL 3p25, MET 7q31, FH 1q42, FLCN 17p11.
> - Lynch (MSH2) → UTUC; BRCA2 → aggressive prostate cancer.

Rapid Revision Sheet – 2021-Summer Paper 2

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

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- Steinstrasse risk rises with burden >2 cm^2; no routine stent.

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- Kidneys en bloc last; Carrel patch, IVC for right vein; machine perfusion reduces DGF.

Q3a. Post prostatectomy incontinence. (SAQ)

- Wait 6–12 months with PFMT before surgery.
- Sling for mild–moderate; AUS for severe or irradiated.
- MASTER: sling non-inferior to AUS at 12 months.

Q3b. Early versus delayed VVF repair. (SAQ)

- Early repair for clean iatrogenic VVF; closure comparable to delayed.
- Delay ~3 months obstetric, ≥12 months radiation.
- Catheter trial works only if small, fresh and started within 3 weeks.

Q3c. Perineal urethrostomy. (SAQ)

- PU bypasses, rather than reconstructs, the diseased anterior urethra.
- Ideal for LS, failed repairs and elderly patients; patient voids sitting.
- BMG PU prevents circumferential scarring; calibrate to 30 Fr.

Q3d. Medical expulsive therapy in urinary calculi. (SAQ)

- Tamsulosin 0.4 mg; best for distal stones 5–10 mm.
- SUSPEND 2015: no benefit in an unselected population.
- Trial of passage 4–6 weeks, then intervene.

Q3e. TOT versus TVT in SUI. (SAQ)

- TOMUS: 80.8% vs 77.7%, equivalent at 1 year.
- TOT: less bladder injury (0.6% vs 4.5%), more groin pain (6.4% vs 1.3%).
- Prefer retropubic TVT in ISD or fixed urethra.

Q3f. Hereditary urological malignancies. (SAQ)

- All familial RCC is autosomal dominant; 3 cm rule except HLRCC.
- VHL 3p25, MET 7q31, FH 1q42, FLCN 17p11.
- Lynch (MSH2) → UTUC; BRCA2 → aggressive prostate cancer.